PRASUN KUMAR

Senior Application Developer, IBM | Agentic AI / GenAI / RAG Engineer | AI/ML Architecture

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PROFESSIONAL SUMMARY

AI/ML and software engineer with 9+ years of overall engineering experience, specializing in Generative AI, Agentic AI, Retrieval-Augmented Generation (RAG), and production AI applications. Hands-on with LangGraph, LangChain, Python, Azure OpenAI, Amazon Bedrock, vector search, LLM evaluation, and cloud-native AI delivery. Experience designing multi-agent workflows, enterprise knowledge assistants, AI document extraction, and LLMOps patterns; previous technical leadership across architecture, code reviews, and stakeholder delivery. Currently Senior Application Developer at IBM (joined 03 Nov 2026).

CORE AI & ENGINEERING SKILLS

Agentic AI & LLMs: LangGraph, LangChain, CrewAI, AutoGen, tool calling, agent state, memory, human-in-the-loop approval, prompt engineering, Azure OpenAI, Amazon Bedrock, Qwen, Llama, Gemma.

RAG & Retrieval: Azure AI Search, FAISS, Chroma, embeddings, BM25, hybrid search, reranking, metadata filters, grounded generation, retrieval and answer evaluation.

AI/ML & Fine-Tuning: scikit-learn, PyTorch, TensorFlow, Hugging Face, Sentence Transformers, NLP classification, LoRA/QLoRA, model benchmarking.

LLMOps & Cloud AI: Databricks, MLflow, evaluation gates, prompt/version tracking, observability, Azure Functions, Azure Service Bus, Azure AI Document Intelligence, AWS Lambda/SQS.

Languages & Platforms: Python, FastAPI, SQL, REST APIs, Docker, Kubernetes, Git, CI/CD, PostgreSQL, MySQL, Azure, AWS.

PROFESSIONAL EXPERIENCE

Senior Application Developer, Band 07 | IBM India Pvt. Ltd. · Pune | Nov 2026 – Present

- Joined IBM India in Pune as a Senior Application Developer (Band 07) on 03 Nov 2026.

Associate AI Specialist (G3A) | Harbinger Group · Pune | Jul 2026 – Nov 2026

- Contributed to GenAI solution design, enterprise RAG, AI-powered automation testing, document workflows, and LLM evaluation patterns.

Associate Technical Lead | Harbinger Group | Jul 2024 – Jun 2026

- Led architecture discussions, design reviews, code quality, mentoring, and delivery planning for AI/ML and Generative AI initiatives.
- Designed AI document-processing systems using Azure OpenAI, Azure AI Document Intelligence, Azure Functions, and Azure Service Bus.

Senior Software Engineer | Harbinger Group | Apr 2021 – Jun 2024

- Developed Python services and machine-learning, NLP, and intelligent automation components for enterprise applications.

Software Professional | Extentia Information Technology | Nov 2019 – Apr 2021

- Built applied NLP prototypes including conversational assistants, resume parsing, semantic skill matching, and early computer-vision experiments.

Associate IT Applications Specialist / Master Level Intern | Symantec | Jan 2017 – Oct 2019

- Established enterprise software engineering foundations; full-time Associate IT Applications Specialist from Jul 2017.

SELECTED AI / GENERATIVE AI PROJECTS — PRIOR ROLES

Agentic AI & Multi-Agent Workflow Automation | LangGraph, LangChain, Python, tool calling

Designed agent workflows with planner, retriever, validator, and executor stages. Applied task routing, state/memory management, retrieval tools, human approval gates, retries, and auditable execution for enterprise use cases. Related public reference: agentic-ai-langgraph-workflows.

Enterprise RAG & Requirements Discovery Agent | Azure AI Search, BM25, embeddings, reranking

Built approaches for document ingestion, chunking, metadata enrichment, semantic and keyword retrieval, and grounded Q&A. Used hybrid retrieval, reranking, answer-quality checks, and evaluation to improve relevance and reduce unsupported answers. Public reference: enterprise-rag-quality-lab.

GenAI Email-to-Case & Document Intelligence | Azure OpenAI, OCR, Azure AI Document Intelligence

Designed AI-driven extraction and validation of email bodies and attachments into structured business records, with schema checks, confidence handling, and human review. Prior project results included a 60%+ improvement in extraction accuracy. Public reference: email-to-case-genai-automation.

AI-Powered Test Automation & Gap Assessment | Amazon Bedrock, LLMs, Databricks, MLflow

Worked on LLM-assisted test generation, document-to-test workflows, evaluation, and environment-aware Azure/AWS AI execution. Explored few-shot prompting, structured outputs, model comparisons, and MLflow-tracked quality metrics. Public reference: ai-test-automation-copilot.

Behavioral Intelligence & LLMOps | Qwen, Llama, Gemma, QLoRA, Databricks

Worked on behavior and NLP classification, open-model evaluation, parameter-efficient fine-tuning, adapter workflows, threshold calibration, and experiment tracking. Focused on model quality and reproducible evaluation for production use.

AI Document Intelligence & Claims Extraction | OCR, LLM extraction, PII/PHI controls

Created document-AI patterns for extraction from scanned PDFs and images, with sensitive-data masking, schema validation, confidence-based review, and auditability. Related public reference: document-intelligence-llm-pipeline.

SELECTED PUBLIC AI REPOSITORIES

github.com/Prasun0512 | agentic-ai-langgraph-workflows • enterprise-rag-quality-lab • mlflow-mlops-ai-platform • ai-jailguard • production-llm-platform-on-kubernetes

Note: Public GitHub repositories are reference implementations/demos; confidential client source code is not published.

EDUCATION, CREDENTIALS & RECOGNITION

MCA: Savitribai Phule Pune University, 2017. | BCA: Sikkim Manipal University, 2014.

Post Graduate Program: Machine Learning & Artificial Intelligence — credential.net/20453200-981b-4985-9ddf-2a44c5ee66cf

Recognition: Technical/team awards and hackathon recognition (2017).